Data Generation Machinery for N-of-1 Clinical Trial Simulations: Appendix B

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This document reproduces Appendix B of the manuscript `report.Rmd`. Cross-references to numbered sections and to Appendix A refer to that manuscript.

Appendix B. Correlation structure of the covariance-moderation and published data-generating processes

This appendix sets out the correlation matrix that each covariance-based data-generating process assembles, so that the differences discussed in Section 2.2.4 can be read off directly. Throughout, `covar` denotes Architecture B as defined in Section 2.2.2 and `orig` denotes the published implementation at commit `58b32a9`. Symbols follow Section 2.1, with the participant subscript i suppressed because every participant's trajectory is drawn from the same matrix: B is the biomarker B_i, and BR_t is the biological response component BR_{it} at occasion t. The vector of biomarker-response correlations is written r, with entries r_t, so that b_i keeps its meaning of the standardized biomarker. Mean moderation does not appear below, because it writes no biomarker-response correlation into the covariance at all; its vector r is identically zero and its interaction is carried in the mean instead.

B.1 Variable ordering and block partition

Both implementations draw each participant's trajectory from a single multivariate normal over the same ordered variable list. With n measurement occasions at cumulative weeks $w_1 < w_2 < \cdots < w_n$, the vector is

$$X = (\, B,\ \mathrm{BL},\ TV_1, \ldots, TV_n,\ PB_1, \ldots, PB_n,\ BR_1, \ldots, BR_n \,)^\top, \qquad \dim X = 2 + 3n.$$

The ordering is factor-major: all n occasions of one factor are contiguous, rather than all three factors at one occasion. Writing A_c for the $n \times n$ within-factor block of factor $c \in \{tv, pb, br\}$ and $C_{cc'}$ for the $n \times n$ cross-factor block, the correlation matrix partitions as

$$R = \begin{pmatrix} 1 & 0 & 0^\top & 0^\top & r^\top \\ 0 & 1 & 0^\top & 0^\top & 0^\top \\ 0 & 0 & A_{tv} & C_{tv,pb} & C_{tv,br} \\ 0 & 0 & C_{pb,tv} & A_{pb} & C_{pb,br} \\ r & 0 & C_{br,tv} & C_{br,pb} & A_{br} \end{pmatrix}.$$

Two features hold in both implementations. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker B couples only to the response factor BR, through the vector $r \in \mathbb{R}^n$; its correlations with TV and PB are zero. The two implementations differ in the interior of A_c, in the interior of $C_{cc'}$, and in r.

B.2 Within-factor blocks A_c

Let ρ_c denote the autocorrelation parameter of factor c (`c.tv`, `c.pb`, `c.br` in the code). Under compound symmetry the same code parameter is a common correlation at every lag and is written $\rho_{cs,c}$; under AR(1) it is a correlation per week and keeps the name ρ_c (Notation, 2026-10-09: ρ_{cs} is the canonical compound-symmetry correlation).

`orig`: compound symmetry. A single constant is assigned to every pair of occasions, irrespective of their separation:

$$[A_c]_{ij} = \begin{cases} 1, & i = j \\ \rho_{cs,c}, & i \neq j \end{cases} \qquad \Longleftrightarrow \qquad A_c = (1 - \rho_{cs,c}) I_n + \rho_{cs,c} J_n,$$

with J_n the $n \times n$ matrix of ones. For $n = 4$,

$$A_c^{\mathrm{orig}} = \begin{pmatrix} 1 & \rho_{cs,c} & \rho_{cs,c} & \rho_{cs,c} \\ \rho_{cs,c} & 1 & \rho_{cs,c} & \rho_{cs,c} \\ \rho_{cs,c} & \rho_{cs,c} & 1 & \rho_{cs,c} \\ \rho_{cs,c} & \rho_{cs,c} & \rho_{cs,c} & 1 \end{pmatrix}.$$

The block has two distinct eigenvalues, $1 + (n-1)\rho_{cs,c}$ once and $1 - \rho_{cs,c}$ with multiplicity $n-1$, so positive definiteness of the block requires $-1/(n-1) < \rho_{cs,c} < 1$. We note that it is the lower bound that moves with n, and it binds only for negative $\rho_{cs,c}$. At the positive values used throughout this paper the admissible range of $\rho_{cs,c}$ is not affected by the number of occasions.

Compound symmetry does restrict the moderation parameter c_{bm} more tightly than AR(1) does, but not by this route. The mechanism is given in B.7.

`covar`: AR(1) in calendar time. The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c^{\text{covar}} = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks $4, 8, 9, 10, 11, 12, 16, 20$, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

B.3 Cross-factor blocks $C_{cc'}$

Let c_1 denote the same-occasion cross-factor correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-factor correlation (`c.cfct`).

`orig`. Both are constants:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times, & i \neq j \end{cases} \qquad \Longleftrightarrow \qquad C_{cc'} = c_\times J_n + (c_1 - c_\times) I_n.$$

`covar`. The same-occasion entry is unchanged, but the different-occasion entries inherit the AR(1) decay:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times \, \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

B.4 Biomarker coupling vector r

The vector r carries the biomarker-treatment interaction. Both implementations write it in terms of the same quantity, the residual fraction of the on-drug response at occasion t,

$$\phi_t = \begin{cases} (1/2)^{t_{sd,t}/t_{1/2}}, & \text{occasion } t \text{ off drug, after first exposure, } t_{1/2} > 0 \\ 0, & \text{otherwise,} \end{cases}$$

where $t_{sd,t}$ is the time since discontinuation. ϕ_t is zero throughout when $t_{1/2} = 0$ and at off-drug occasions before first exposure, and it is the factor by which carryover multiplies the on-drug response mean.

`covar`. The entry equals c_{bm} on drug and is scaled by the residual fraction off drug:

$$r_t = \begin{cases} c_{bm}, & \text{occasion } t \text{ on drug} \\ c_{bm}\,\phi_t, & \text{occasion } t \text{ off drug.} \end{cases}$$

Since $\phi_t = e^{-\lambda t_{sd,t}}$ with $\lambda = \ln 2 / t_{1/2}$, this is the exponential decay of Section 2.2.2.

`orig`. The entry equals c_{bm} wherever the response mean $\mu_{BR,t}$ is nonzero and zero elsewhere. Because $\mu_{BR,t}$ is nonzero off drug exactly when $\phi_t > 0$, this is

$$r_t = \begin{cases} c_{bm}, & \text{occasion } t \text{ on drug} \\ c_{bm}\,\mathbf{1}[\phi_t > 0], & \text{occasion } t \text{ off drug.} \end{cases}$$

The two vectors are therefore nonzero at the same occasions. The published code's condition on $\mu_{BR,t}$ and `covar`'s branch on the drug indicator switch the channel on in the same places, and they differ only in how large the off-drug entry is: graded in proportion to the residual effect under `covar`, and a step to the full on-drug value under `orig`. At $t_{1/2} = 0$, $\phi_t = 0$ at every off-drug occasion and the two vectors are equal entry for entry. For $t_{1/2} > 0$ both sets of off-drug entries rise above zero, but the `covar` entries rise only as far as the residual fraction allows, while the `orig` entries jump to c_{bm}. In a design whose every path begins on drug, such as the Hybrid design, $r^{\text{orig}} = c_{bm}\mathbf{1}_n$ at any positive half-life, so the vector carries no on-drug versus off-drug contrast at all.

B.5 A worked example

To make the three components concrete, take four occasions drawn from the Hybrid design at cumulative weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_\times = 0.1$, $c_{bm} = 0.45$,

and a carryover half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$(|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

The unequal spacing matters: occasions one to three are a week apart, while the fourth sits four weeks after the third.

B.5.1 Within-factor block A_{br}

$$A_{br}^{\text{orig}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.6000 & 0.6000 \\ 0.6000 & 1.0000 & 0.6000 & 0.6000 \\ 0.6000 & 0.6000 & 1.0000 & 0.6000 \\ 0.6000 & 0.6000 & 0.6000 & 1.0000 \end{pmatrix}, \qquad A_{br}^{\text{covar}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1.0000 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1.0000 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1.0000 \end{pmatrix}.$$

Under `orig` every pair of occasions carries 0.6, so the week-apart pair $(1, 2)$ and the six-weeks-apart pair $(1, 4)$ are treated as equally associated. Under `covar` the same two pairs carry 0.6 and 0.0467, a factor of thirteen apart. The consequence for the fourth occasion is the striking one: it is nearly independent of the first three under AR(1), and as strongly tied to them as they are to each other under compound symmetry.

B.5.2 Cross-factor block $C_{tv,pb}$

$$C_{tv,pb}^{\text{orig}} = \begin{pmatrix} 0.3000 & 0.1000 & 0.1000 & 0.1000 \\ 0.1000 & 0.3000 & 0.1000 & 0.1000 \\ 0.1000 & 0.1000 & 0.3000 & 0.1000 \\ 0.1000 & 0.1000 & 0.1000 & 0.3000 \end{pmatrix}, \qquad C_{tv,pb}^{\text{covar}} = \begin{pmatrix} 0.3000 & 0.0400 & 0.0160 & 0.0004 \\ 0.0400 & 0.3000 & 0.0400 & 0.0010 \\ 0.0160 & 0.0400 & 0.3000 & 0.0026 \\ 0.0004 & 0.0010 & 0.0026 & 0.3000 \end{pmatrix}.$$

The diagonal, which carries the same-occasion cross-factor correlation c_1, is identical in both. The off-diagonal entries decay under `covar` and do not under `orig`, and by the fourth occasion the difference is two orders of magnitude.

A caution attaches to the `covar` block. In the implementation the decay rate applied to a cross-factor pair is taken from whichever factor the outer loop is currently visiting, and because each unordered pair is visited in both orders, the later write survives. The rate appearing in $C_{tv,pb}$ is therefore determined by loop order rather than by any modeling decision. The values above use ρ_{pb} on that basis. This is an implementation artifact and not a property of the AR(1) specification.

B.5.3 Biomarker vector r

At a carryover half-life of one week,

$$r^{\text{covar}} = (0.4500,\, 0.2250,\, 0.1125,\, 0.4500)^\top, \qquad r^{\text{orig}} = (0.4500,\, 0.4500,\, 0.4500,\, 0.4500)^\top.$$

The two off-drug entries halve and quarter under `covar`, tracking $\phi_t = (1/2, 1/4)$, whereas under `orig` they take the full c_{bm}, because carryover has left the response mean nonzero and the step therefore returns the full on-drug value.

Removing carryover collapses the difference entirely:

$$r^{\text{covar}} = r^{\text{orig}} = (0.4500,\, 0.0000,\, 0.0000,\, 0.4500)^\top \qquad (t_{1/2} = 0).$$

This pair of displays is the essential comparison. The vector r is the only channel through which the biomarker-treatment interaction enters the covariance, and an interaction is identified by r varying between on-drug and off-drug occasions. At $t_{1/2} = 0$ the two implementations produce the same vector. At any positive half-life r^{orig} is constant, so it carries no on-drug versus off-drug contrast, while r^{covar} retains one. Across the grid $t_{1/2} \in \{0, 0.5, 1.0\}$ used in this paper, `orig` therefore supplies one cell identical to `covar` and two in which its interaction channel is degenerate.

B.6 The three data-generating processes

The table below lists only the entries in which the three processes differ, grouped by what each group governs. Rows in which all three agree are recorded in prose beneath it, so that the table can be read as a list of differences rather than a full specification.

	`mean`	`covar`	`orig`
Where the interaction is written			
Moment used	μ	Σ	Σ
Correlation structure			
A_c within-factor	$\rho_c^{\lvert w_t - w_s \rvert}$	$\rho_c^{\lvert w_t - w_s \rvert}$	$\rho_{cs,c}$ (constant)
$C_{cc'}$ off-diagonal	$c_\times \rho^{\lvert w_t - w_s \rvert}$	$c_\times \rho^{\lvert w_t - w_s \rvert}$	$c_\times$ (constant)
Interaction channel			
r_t, on drug	0	c_{bm}	c_{bm}
r_t, off drug	0	$c_{bm}\, \phi_t$	$c_{bm}\, \mathbf{1}[\phi_t > 0]$
μ_{BR} shift, on drug	$\beta_{bm} b_i \sigma_{BR}$	none	none
Carryover on the BR mean			
Decay anchored at	μ_{BR,t^*}	μ_{BR,t^*}	$\mu_{BR,t-1}$

Notation. μ and Σ are the mean vector and covariance matrix of the multivariate normal from which each participant's trajectory is drawn. c indexes the three response factors, $c \in \{tv, pb, br\}$, and t indexes measurement occasions, which fall at cumulative weeks $w_1 < \cdots < w_n$.
A_c is the $n \times n$ within-factor block of factor c, giving the correlation between that factor at two occasions. $C_{cc'}$ is the $n \times n$ cross-factor block between two distinct factors. Its diagonal holds same-occasion entries and its off-diagonal different-occasion entries.
ρ_c is the within-factor autocorrelation parameter of factor c (`c.tv`, `c.pb`, `c.br` in the code), written $\rho_{cs,c}$ when it is the common correlation of compound symmetry. ρ without a subscript is the corresponding decay rate applied to cross-factor entries. c_1 is the same-occasion cross-factor correlation (`c.cf1t`) and $c_\times$ the different-occasion one (`c.cfct`). The exponent $|w_t - w_s|$ is the gap in cumulative weeks between occasions t and s, not the difference in occasion index.
r_t is the correlation between the biomarker and the response factor at occasion t, the vector through which covariance moderation carries the interaction. c_{bm} is the moderation parameter, $t_{sd,t}$ the time since drug discontinuation, and $t_{1/2}$ the carryover half-life. $\phi_t = (1/2)^{t_{sd,t}/t_{1/2}}$ is the residual fraction of the on-drug response at an off-drug occasion that follows drug exposure, and $\phi_t = 0$ otherwise, including throughout when $t_{1/2} = 0$ and at off-drug occasions before first exposure. `covar` and `orig` are nonzero at the same occasions; they differ in whether r_t scales with the residual fraction or steps to its full on-drug value.
μ_{BR} is the response factor's mean trajectory and $b_i = (B_i - \mu_{bm})/\sigma_{bm}$ the participant's standardized biomarker, so the mean-moderation shift $\beta_{bm} b_i \sigma_{BR}$ is expressed per standard deviation of the biomarker. t^* denotes the most recent on-drug occasion and $t - 1$ the immediately preceding occasion, which is the whole of the difference in the final row: all three processes apply the same factor $(1/2)^{t_{sd}/t_{1/2}}$ and differ only in what they apply it to.
BL is baseline symptom severity, B the biomarker, and TV, PB, BR the natural-history, placebo-belief and pharmacological response factors respectively.

Five entries agree across all three processes and are omitted above. The same-occasion cross-factor correlation is c_1 throughout. The baseline row and column are those of the identity. The biomarker correlates with neither the natural-history nor the placebo-belief factor. The off-drug μ_{BR} shift is absent everywhere, since mean moderation applies its shift only at on-drug occasions. And all three apply the same exponential decay factor $(1/2)^{t_{sd}/t_{1/2}}$, differing only in the value it multiplies, which is the last row of the table.

Three features of this comparison are worth drawing out.

First, `mean` and `covar` share their entire covariance matrix apart from the vector r. Setting r to zero and adding the mean shift converts one into the other, with every within-factor and cross-factor block untouched. This is the decomposition of Appendix A made concrete: the two architectures are not different models but the two places a single joint distribution can carry an interaction.

Second, `orig` differs from `covar` on three rows simultaneously, in A_c, in the off-diagonal of $C_{cc'}$, and in r. No single row can therefore be held responsible for any difference in results obtained under it, and a comparison against `orig` bounds the joint effect of the three differences rather than decomposing them. Separating them would require running the correlation structure and the off-drug coupling rule as crossed factors. The carryover row is a fourth difference, but it is of our making rather than Hendrickson's, and Section B.7 explains why it should be held fixed across arms rather than counted among the architectural differences.

Third, the off-drug r_t row is the consequential one, and it is consequential because of how r_t is sized, not where it is nonzero. The two covariance-based processes switch the interaction

channel on at exactly the same occasions: on drug, and off drug after exposure whenever $t_{1/2} > 0$. We confirmed this against the package's own matrix construction on every path of the three designs at each half-life in the grid. They differ in what happens there. Under `covar` the correlation is scaled by the residual fraction ϕ_t, so that it falls in proportion to the residual drug effect in the response mean. Under `orig` it steps to the full c_{bm} as soon as any residual effect is present, however small. With carryover, the published coupling is therefore constant at every post-exposure occasion, and its interaction channel loses the on-drug versus off-drug contrast on which identification depends.

The carryover row requires explanation, since the three processes agreed on it until recently and a reader comparing code will want to identify the difference quickly.

All three adjust the response mean at an off-drug occasion by adding a decayed fraction of the previous response. They differ in what is decayed. Writing t^* for the most recent on-drug occasion,

$$\underbrace{\mu_{BR,t} + \mu_{BR,t^*}\,(1/2)^{t_{sd,t}/t_{1/2}}}_{\texttt{mean},\ \texttt{covar}} \qquad \text{against} \qquad \underbrace{\mu_{BR,t} + \mu_{BR,t-1}\,(1/2)^{t_{sd,t}/t_{1/2}}}_{\texttt{orig}}\,.$$

The published form multiplies the *already adjusted* mean at $t-1$ while $t_{sd,t}$ is measured cumulatively from discontinuation, so elapsed time is counted twice. The corrected form applies the cumulative factor once, to the mean at discontinuation.

Two consequences make the difference easy to spot in output. The first off-drug occasion of any run agrees exactly under both forms, since there is nothing yet to inherit. Divergence begins at the second. In the worked example of Section B.5, at $t_{1/2} = 1$ week, occasion 3 carries $0.25\,\mu_{BR,1}$ under the corrected form and $0.125\,\mu_{BR,1}$ under the published one, a factor of two. The gap widens with each further off-drug occasion, so designs with long uninterrupted washouts show it most.

We corrected `mean` and `covar` in September 2026 and leave `orig` as published, since the object of that column is to describe the published code rather than to improve it.

The consequence for the re-run flagged in Section 2.2.4 should be settled before that run is designed. A comparison of `covar` against `orig` now carries one difference beyond the four enumerated there, and it is of our making. Running `covar` as corrected asks what the architecture delivers today and credits the correction to `covar`. Running `covar` with the recursion restored isolates the four architectural differences and answers the question Section 2.2.4 poses. We prefer the second for the headline comparison, with the first reported alongside it, so that the cost of the correction remains visible.

B.7 Summary of differences

Component	`orig`	`covar`
A_c interior	$\rho_{cs,c}$	$\rho_c^{\lvert w_t - w_s \rvert}$
$C_{cc'}$, $t \neq s$	$c_\times$	$c_\times \rho^{\lvert w_t - w_s \rvert}$
r_t, off drug	$c_{bm}\,\mathbf{1}[\phi_t > 0]$	$c_{bm}\,\phi_t$
Carryover on μ_{BR}	recursive	anchored

The table lists only the entries that differ. Everything else in the correlation matrix agrees: the same-occasion cross-factor entry c_1, the on-drug value c_{bm} of the coupling vector, the BL row and column, which are those of the identity in both processes, and the B–TV and B–PB entries, which are zero in both. That last agreement is worth stating rather than leaving implicit, since it is the assumption that the biomarker couples to no factor other than the response, and a good deal rests on it.

The four differences are independent decisions rather than consequences of one another, and they are not of equal standing. Only one of them repairs a defect. We set out what each is for, since a reader is entitled to ask whether a departure from published code was necessary or merely preferred.

The off-drug coupling (r_t off drug) corrects a defect. This is the one substantive repair. Both processes switch the coupling on at the same occasions, so the defect is not where the published code opens the channel but how far. Under the published rule the off-drug correlation takes the full on-drug value c_{bm} at any occasion carrying residual drug effect, however small that residual is. The coupling is therefore a step function of the residual effect rather than a function of its size: an off-drug occasion at which the response mean has fallen to a few percent of its on-drug value is assigned the same biomarker-response correlation as an on-drug occasion. In a design whose paths all begin on drug, the coupling vector is then constant at every positive half-life, and the interaction channel carries no on-drug versus off-drug contrast at all. Since the interaction is identified by precisely that contrast, the introduction of any carryover at all removes it, which inverts the graded effect residual exposure should have. Architecture B instead scales the coupling by the residual fraction ϕ_t, so that the biomarker-response correlation falls in proportion to the residual drug effect in the response mean. The ceiling discussion below shows that the published rule admits a wider nominal c_{bm} only because its vector has stopped carrying an interaction, which is a further symptom of the same defect rather than a separate finding.

The repository was itself revised in 2024, replacing the step with a graded coupling. We do not offer that revision as independent corroboration. The commit in question (`8609f12`, 6 May 2024) is authored by Hendrickson but its message describes it as the present author's edits curated into working code, so the two lines of reasoning are not independent. The argument for the graded form rests on the identification point above, not on the repository's later history.

The within-factor form (A_c) is a modeling choice. It is adopted for clinical plausibility, since nearby measurements should be more strongly correlated than distant ones, and for the wider feasible range discussed below. It repairs nothing. Compound symmetry is a defensible model of within-participant correlation, and it is the standard implication of a participant random intercept.

The cross-factor form ($C_{cc'}$, $i \neq j$) follows from the previous row. It is not listed among the four differences of Section 2.2.4, and on its own it would be difficult to defend. Taken without the AR(1) change it lowers the feasible range of c_{bm} substantially, which is the numerical signature of an incoherent structure: a constant cross-factor term alongside a decaying within-factor block asserts that a factor's correlation with a *different* factor at eight occasions' remove equals its correlation with that factor at the adjacent occasion, while its correlation with *itself* has decayed away. `covar` takes the two rows together for consistency, not because either entails the other.

The carryover row is ours, not Hendrickson's. All three processes originally adjusted the response mean by the recursive form, and we corrected `mean` and `covar` in September 2026 for the reason given in Section B.6. The row is listed here because a reader comparing code will encounter it, but it should not be counted among the architectural differences that Section 2.2.4 enumerates.

The feasible range of c_{bm}. The two forms differ in how much moderation the covariance can carry. The mechanism is worth setting out, because the obvious explanation is wrong and the direction of the difference is not constant.

The binding constraint is not positive definiteness of the within-factor block. It is a Schur complement on the biomarker row, and we set the argument out in full because it replaces a grid search with a formula.

Drop the BL row and column, which are those of the identity and therefore factor out of every determinant. Order the remainder with the biomarker first and partition accordingly:

$$R = \begin{pmatrix} 1 & \tilde{r}^\top \\ \tilde{r} & M \end{pmatrix},$$

where M is the $3n \times 3n$ block of the three response factors and $\tilde{r}$ is the biomarker coupling vector extended by zeros, equal to r on the BR positions and vanishing on the others. The Schur complement of M in R is the scalar $1 - \tilde{r}^\top M^{-1} \tilde{r}$, and R is positive definite if and only if M is positive definite and that scalar is positive.

The scalar has a statistical reading which makes the constraint intuitive rather than merely algebraic. On the correlation scale the biomarker has unit variance. The quantity $\tilde{r}^\top M^{-1} \tilde{r}$ is the coefficient of determination from regressing the biomarker on the whole set of response variables, and $1 - \tilde{r}^\top M^{-1} \tilde{r}$ is the residual variance of the biomarker once that regression

is removed. The condition therefore says only this: the biomarker may not be more than perfectly predictable from the response factors. An analyst who specifies correlations strong enough to imply an R^2 above one has not specified a covariance matrix at all, having asked for a negative variance.

Writing $\tilde{r} = c_{bm} v$, where v carries the on-drug and off-drug pattern of Section B.4 with the moderation parameter factored out, the condition becomes $c_{bm}^2 \, v^\top M^{-1} v < 1$ and the exact ceiling follows:

$$c_{bm}^{*} = \left(v^\top M^{-1} v \right)^{-1/2}.$$

The bound is attained rather than approached. At $c_{bm} = c_{bm}^{*}$ the implied R^2 is exactly one, the residual variance is exactly zero, and the smallest eigenvalue of R reaches zero at the same point. We have verified this correspondence directly on small cases, where the three quantities can be watched together as c_{bm} is increased.

Two features of this expression matter. The ceiling depends on the correlation structure only through how the coupling pattern v sits relative to M^{-1}, and it depends on v, which differs between the two processes and, under `orig`, changes with the carryover half-life.

That second point reverses the comparison. At $t_{1/2} = 0$ the `orig` coupling is the on-drug indicator and its ceiling is the lower of the two. At any positive half-life the `orig` coupling becomes constant, as Section B.4 shows, and a constant v aligns with the direction in which M is best conditioned. Its ceiling is then the higher of the two. The apparent gain is worthless: it arises precisely because the vector has lost the on-drug versus off-drug contrast on which identification depends. A process that can carry a larger nominal c_{bm} while carrying no interaction is not thereby preferable.

We note that the number of occasions is not what separates the two forms. Both ceilings fall as occasions are added, in similar proportion. The smallest eigenvalue of the compound-symmetric block M is $(1 - \rho_{cs,c}) - (c_1 - c_\times)$, which contains neither n nor the number of factors. An earlier version of this appendix attributed the difference to a constraint on $\rho_{cs,c}$ that tightens with n; that attribution was incorrect.

[TO RE-VERIFY: the relations above were derived analytically and confirmed on an evenly spaced schedule. Design-specific thresholds for the CO, Hybrid and OL+BDC schedules of Section 2.3 have not been computed, and none is quoted here. Evaluate c_{bm}^{*} on those three designs, under `58b32a9` for `orig`, before restoring numeric thresholds.]

B.8 The mean-covariance contrast

Sections B.6 and B.7 compare Architecture B against the published implementation. We give the corresponding comparison for the contrast this paper actually reports, mean moderation against covariance moderation, because it is much the cleaner of the two.

Where the interaction is written. The two processes place the interaction in different objects, and the first table records only that.

Component	`mean`	`covar`
r_t, on drug	0	c_{bm}
r_t, off drug	0	$c_{bm}\,\phi_t$
μ_{BR} shift, on drug	$\beta_{bm} b_i \sigma_{BR}$	none

Everything else is shared. Both processes use the AR(1) within-factor form, the same cross-factor blocks, the same anchored carryover adjustment, the same baseline row and column, and the same zero correlations between the biomarker and the two non-response factors. Setting r to zero and adding the mean shift converts one process into the other, and nothing else moves.

A caution about the zero. The entry $r_t = 0$ describes the covariance matrix from which the data are drawn. It does not say that the biomarker and the response are uncorrelated under mean moderation. The shift is applied after the draw, and it induces an association of its own. Reading the zero as an absence of biomarker-response dependence would be a mistake, and the second table is included to forestall it.

What each process realizes. Writing σ for σ_{BR}, taking an on-drug occasion, and setting the two parameters to their matched value $\beta_{bm} = c_{bm}$, written c_{bm} in the table, the two processes differ in every second moment while agreeing exactly on the first.

Quantity	`mean`	`covar`
$E[BR_t \mid B_i]$	$\mu_{BR} + c_{bm}\dfrac{\sigma}{\sigma_{bm}}(B_i - \mu_{bm})$	same
Slope of BR_t on B_i	$c_{bm}\sigma/\sigma_{bm}$	same
SD$[BR_t \mid B_i]$	σ	$\sigma\sqrt{1 - c_{bm}^2}$
Marginal SD$[BR]$	$\sigma\sqrt{1 + c_{bm}^2}$	σ
Cor(B_i, BR_t)	$c_{bm}/\sqrt{1 + c_{bm}^2}$	c_{bm}

At $c_{bm} = 0.45$ and $\sigma = 8$ these give a realized correlation of 0.410 under mean moderation against 0.450 under covariance moderation, conditional standard deviations of 8.00 against 7.15, and marginal standard deviations of 8.77 against 8.00. A two-million-draw simulation reproduces each to three decimal places.

Three things follow, and together they are the reason we prefer this comparison to the one against `orig`.

First, the conditional means agree exactly. The mean shift is calibrated to reproduce the conditional expectation that Architecture B's covariance implies, so the two processes present

the same regression slope to the analysis model. Whatever difference Section 3 reports is not a difference in the quantity being estimated.

Second, they reach that common slope by opposite routes, and this is the substantive content of the architecture distinction. Covariance moderation holds the marginal variance fixed and shrinks the conditional variance, so the biomarker explains variation already present in the population. Mean moderation holds the conditional variance fixed and inflates the marginal, so the biomarker introduces new between-patient spread. Appendix A.2 puts the same point qualitatively: a biomarker that indexes an unobserved responder status against one that fixes the drug effect. The dilution of the correlation from 0.450 to 0.410 is the arithmetic residue of the added variance.

Third, the comparison is clean in a way the `orig` comparison is not. Section B.7 lists four differences, and a power difference observed against `orig` cannot be attributed to any one of them without further work. Here there is a single difference, expressed in two complementary rows, and its consequences are the five moments tabulated above.

We note one practical implication. Because the marginal variances differ, absolute power is not comparable between the two architectures at a common nominal c_{bm}, and Section 3 accordingly compares the profile across carryover within each architecture rather than the levels between them.

The two processes do not, however, carry the interaction at the same occasions. Mean moderation applies its shift only where the drug indicator is 1, so its interaction is absent at every off-drug occasion whatever the carryover half-life. Covariance moderation sets $r_t = c_{bm}$ at those same on-drug occasions and also carries a graded interaction, $c_{bm}\phi_t$, at off-drug occasions that follow exposure whenever $t_{1/2} > 0$. The two therefore agree exactly when $t_{1/2} = 0$ and diverge off drug as soon as carryover is present. This is the data-generating counterpart of the point made in Section 2.2.1: the analysis regressor $D_{bc,it}$ has the same graded off-drug profile as ϕ_t, so it matches covariance moderation's off-drug interaction and finds none under mean moderation. The published implementation shares covariance moderation's support but not its scaling, which is the difference Section B.7 identifies as consequential.